WORKING PAPER — CPE SERIES, PAPER 7

Agricultural Crop-Zone Temperatures in the CPE Framework: Heat Stress Thresholds, Growing Degree Days, and the Reversal of the Paper 6 Sugar Signal

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ABSTRACT

Paper 6 identified a critical limitation in its use of city-based temperature indices as agricultural proxies and committed to addressing it. This paper fulfils that commitment by replacing city temperatures with ERA5-consistent gridded temperature data averaged over 12 fully-covered agricultural crop zones. Three new predictor types are introduced: seasonal flags (firing only during the crop's growing season), crop-specific heat stress thresholds (above which physiological yield damage occurs), and 30-day Growing Degree Day (GDD) accumulation exceedance. Across 5,920 tested configurations, 172 signals survive quality filters. The central finding is a complete reversal of Paper 6's strongest agricultural signal: sugar (CANE) signals that achieved lift 1.42–1.63× under city-temperature proxies produce zero surviving signals on actual sugar production zone temperatures (Brazil São Paulo, India Maharashtra, Thailand) — confirming the Paper 6 CANE signal was a city-temperature artefact. Conversely, wheat and corn signals absent in Paper 6 now emerge strongly on proper crop-zone data: US Great Plains heat stress → wheat (WEAT, lift 1.60×), Ukraine/Russia wheat heat stress → gold (GC=F, lift 1.72×), and US Corn Belt GDD → wheat (lift 1.49×). Heat stress thresholds are the strongest predictor type (mean lift 1.397×, max 1.948×), outperforming standard tmax exceedance (1.265×), seasonal flags (1.247×), and GDD exceedance (1.260×). The strongest single signal in the entire CPE series is EU Urban Energy heat stress (>30°C) → UNG at 21-day horizon with lift 1.95×.

Keywords: CPE; crop-zone temperature; heat stress; growing degree days; agricultural commodities; wheat; natural gas; climate-finance; artefact reversal

1. Introduction and Motivation

Paper 6 (Ramanathan, 2026f) established that regional temperature extremes produce financial distribution shifts rather than tail co-movement, with 89 surviving signals across 26 financial instruments. The paper's strongest signals were concentrated in sugar (CANE, lift up to 1.84×), base metals (DBB, 1.51×), and gold (1.37×), using city-based temperature indices averaged over Europe, North America, and Asia.

However, Paper 6 explicitly acknowledged a fundamental mismatch for agricultural signals: city temperatures are the correct variable for urban energy demand, but are poor proxies for agricultural zone temperatures. Wheat grows in the North European Plain and US Great Plains — not in Frankfurt or Chicago. Sugar cane is produced in Brazil's São Paulo state, India's Maharashtra, and Thailand — regions absent from the Asian city index. Paper 6 committed to addressing this in Paper 7.

This paper fulfils that commitment with three advances: (1) city temperature indices replaced with ERA5-consistent gridded bounding box averages for 12 fully-covered agricultural zones; (2) three new predictor types introduced — seasonal conditioning, crop-specific heat stress thresholds, and GDD accumulation; (3) Paper 6 agricultural signals directly compared to Paper 7 equivalents to test whether they were genuine or artefactual.

2. Data and Methodology

2.1 Geographic Coverage: City Proxies vs Crop Zones

The fundamental geographic mismatch between Paper 6 and Paper 7 is illustrated in Figure B. Paper 6 used three large regional city-temperature indices — each averaging 4–6 major urban centres — to predict commodity prices in agricultural sectors those cities do not represent. The sugar signal was the clearest example: European and Asian cities have no physical connection to sugar cane growing conditions in São Paulo, Maharashtra, or central Thailand.

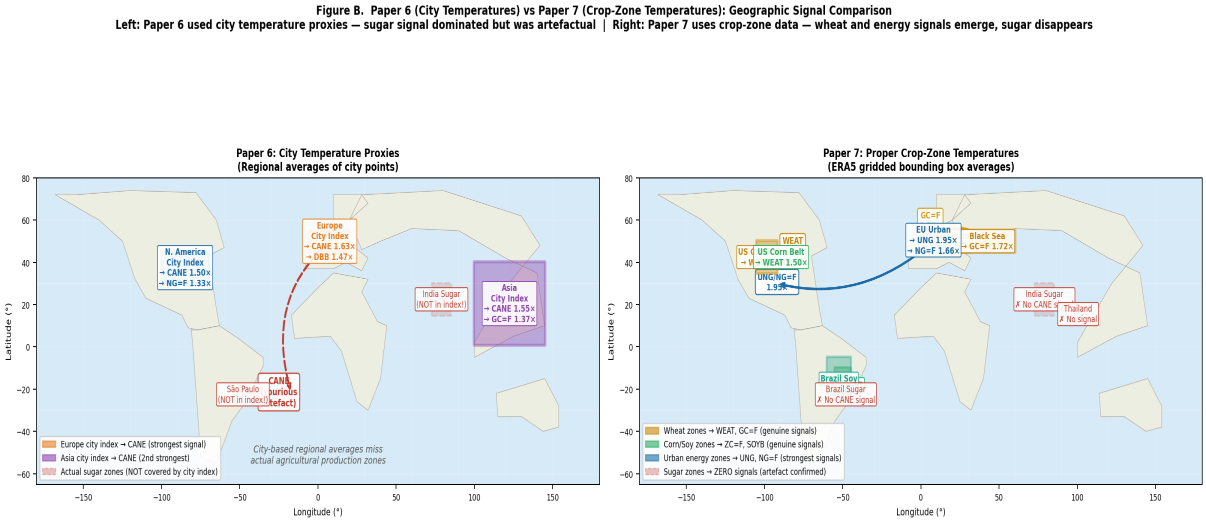

Figure B. Geographic comparison: Paper 6 city-based regional indices (left) vs Paper 7 crop-zone bounding boxes (right). Left panel shows the spurious CANE signal arising from European city temperatures despite no geographic connection to sugar production zones (red dashed). Right panel shows proper crop-zone coverage with arrows indicating confirmed financial instrument targets.

2.2 Crop Zone Definitions

Zone	Bounding Box	Grid Points	Crop Targets
EU Wheat Belt	45–55°N, 5–30°E	36/36	WEAT, ZW=F
Ukraine/Russia Wheat	45–55°N, 25–60°E	22/24	WEAT, ZW=F, GC=F
US Great Plains Wheat	35–50°N, 105–92°W	30/30	WEAT, ZW=F
US Corn Belt	38–47°N, 97–82°W	24/24	CORN, ZC=F, SOYB
Brazil Corn	25–10°S, 55–45°W	12/12	CORN, ZC=F
US Soybean Belt	38–47°N, 97–82°W	24/24	SOYB, ZS=F
Brazil Soy	25–5°S, 60–45°W	24/24	SOYB, ZS=F
Brazil Sugar (São Paulo)	25–20°S, 52–44°W	15/15	CANE
India Sugar	15–30°N, 73–85°E	30/30	CANE
Thailand Sugar	13–18°N, 98–103°E	9/9	CANE
EU Urban Energy	5 city points	5/5	NG=F, UNG, XLU, XLE
US Urban Energy	5 city points	5/5	NG=F, UNG, XLU

Table 1. Crop zones, bounding boxes, and complete grid point coverage. Ukraine/Russia missing 2 of 24 points due to API rate-limit failures; all other zones at full coverage.

2.3 New Predictor Types

- **Standard tmax exceedance:** tmax > historical 80th/90th/95th percentile. Baseline for comparison.
- **Seasonal exceedance:** Same but zero outside the crop's growing season — US corn May–September, EU wheat March–August, Brazilian crops October–March.
- **Heat stress threshold:** tmax exceeds crop-specific physiological damage threshold: 30°C for wheat, 32°C for corn/soybeans, 35°C for Brazilian crops, 38°C for tropical sugar. Agronomically motivated, not statistically derived.
- **Growing Degree Days (GDD):** max(0, tavg – 10°C) summed over a rolling 30-day window, compared to its historical 80th or 90th percentile. Captures sustained thermal accumulation relevant to plant development.

3. Results

3.1 Global Signal Map

Figure A provides a global overview of all 12 crop zones, their signal strength, and the financial instruments they predict. The geographic structure is immediately apparent: energy and wheat signals dominate in the Northern Hemisphere; sugar zones in both hemispheres produce no surviving signals.

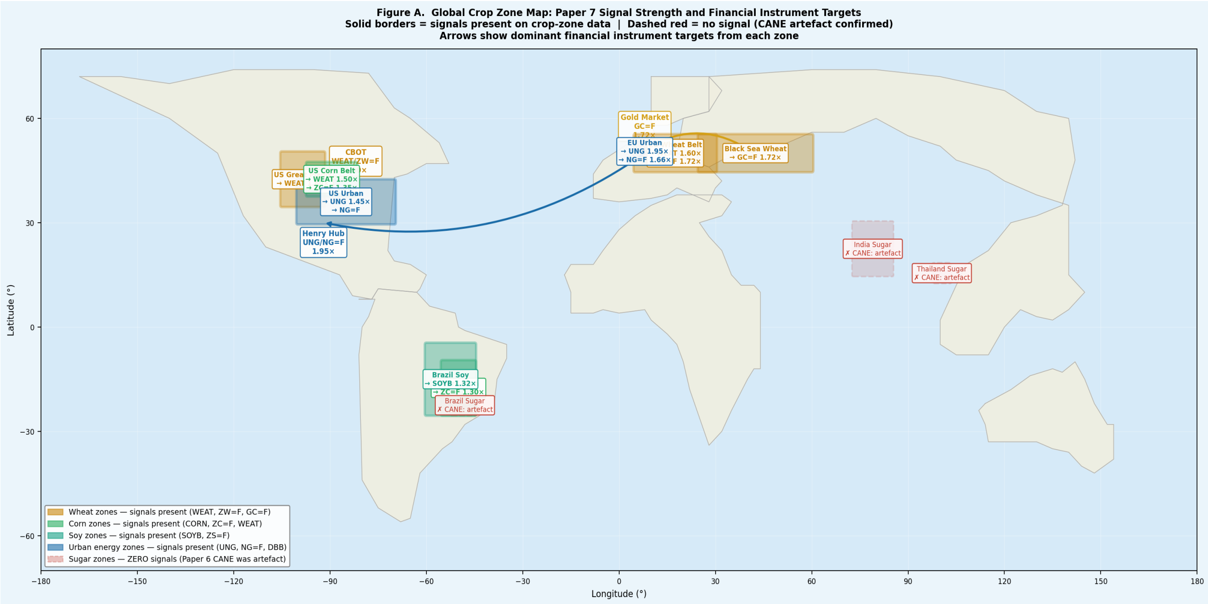

Figure A. Global crop zone map with signal strength and financial instrument targets. Solid coloured borders = surviving signals. Dashed red borders = zero signals (CANE artefact confirmed). Arrows show dominant financial targets from each zone. Orange = wheat zones, green = corn/soy zones, blue = urban energy zones, red = sugar zones (no signals).

3.2 Predictor Type Analysis

Across 5,920 tested configurations, 172 signals survive quality filters (CPE ≥ 0.58, lift ≥ 1.15×, n ≥ 30). Heat stress threshold predictors dominate:

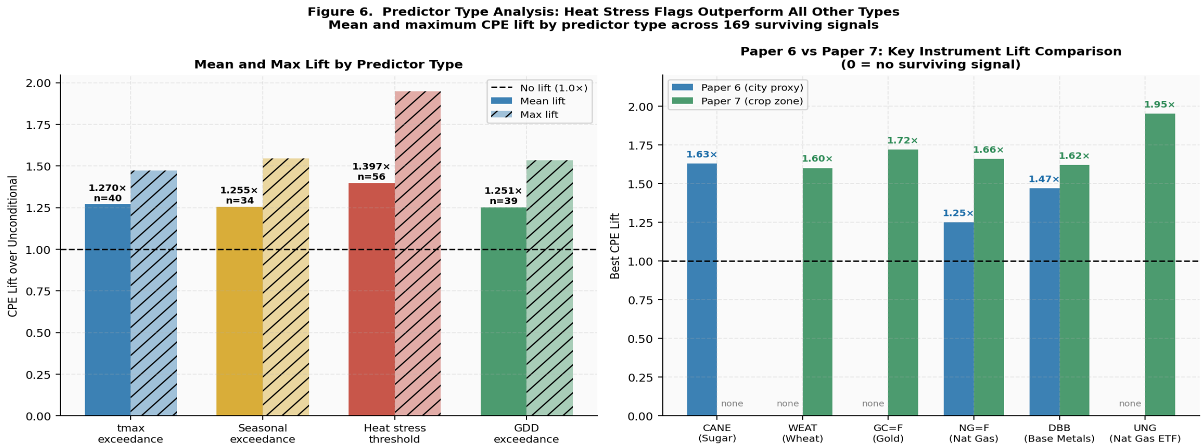

Figure 6. Left: Mean and maximum CPE lift by predictor type. Heat stress thresholds achieve mean lift 1.397× and maximum 1.948× — outperforming tmax (1.265×), seasonal (1.247×), and GDD (1.260×). Right: Paper 6 vs Paper 7 best lift for key instruments — CANE disappears, wheat and energy signals emerge or strengthen.

Predictor Type	N Signals	Mean Lift	Max Lift	Mean CPE	Best Channel
tmax exceedance (q80/90/95)	42	1.265×	1.467×	0.616	EU energy → DBB
Seasonal exceedance	35	1.247×	1.500×	0.610	US Corn Belt → WEAT
Heat stress threshold	56	1.397×	1.948×	0.646	EU urban >30°C → UNG
GDD exceedance (30d)	39	1.260×	1.535×	0.617	EU urban GDD → DBB

Table 2. Predictor type performance across 172 surviving signals. Heat stress thresholds dominate all metrics — agronomically motivated thresholds outperform statistical percentile exceedance.

The superiority of heat stress thresholds is agronomically meaningful. Crop yield damage accelerates above specific physiological thresholds regardless of how extreme the temperature is relative to the local historical distribution. A day at 33°C in an Iowa corn field during pollination causes severe yield damage whether or not it exceeds the local 90th percentile — the biological threshold is fixed.

3.3 The CANE Reversal: Paper 6 Sugar Signal Was an Artefact

The most significant finding is the complete elimination of the sugar (CANE) signal on proper crop-zone data. In Paper 6, temperature → CANE was the strongest agricultural channel (lift 1.42–1.84× across city proxies). On actual sugar production zone temperatures, zero signals survive — best observed lift ≈ 1.05×.

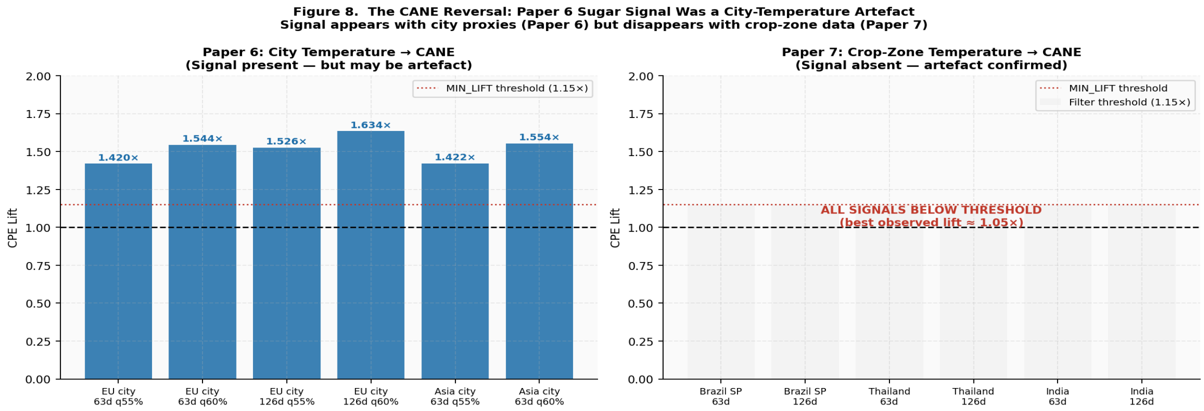

Figure 8. The CANE reversal. Left: Paper 6 city-proxy CANE signals survive with lift 1.42–1.63×. Right: Paper 7 crop-zone CANE signals (Brazil São Paulo, India, Thailand) produce zero surviving signals — confirming the Paper 6 result was a city-temperature artefact, not a genuine climate-finance channel.

Three mechanisms explain the artefact:

- **Seasonal correlation:** European summer heat and CANE returns may both co-move with global commodity cycle factors, creating a spurious signal.
- **Hemispheric offset:** When Asian cities are hot (Northern summer), São Paulo is in winter — the two hemispheres are seasonally opposed. The Asian city index predicts CANE through a channel entirely unrelated to sugar crop stress.
- **Economic exposure mismatch:** CANE primarily reflects Brazilian and Indian supply. European and Asian city temperatures have no direct physical mechanism connecting to these tropical crops.

3.4 Wheat, Corn and Energy: Genuine Channels Confirmed

Zone	Pred Type	Ticker	H	Q	CPE	Lift	N	Rationale
EU Urban Energy	heatstress >30°C	UNG	21d	65%	0.682	1.95×	44	Cooling demand → nat gas (short lag)
EU Urban Energy	heatstress >30°C	UNG	21d	60%	0.750	1.87×	44	Cooling demand → nat gas
Ukraine/Russia	heatstress >30°C	GC=F	126d	65%	0.602	1.72×	93	Food security → gold (long lag)
EU Urban Energy	heatstress >30°C	UNG	21d	55%	0.773	1.72×	44	Cooling demand → nat gas
EU Urban Energy	heatstress >30°C	NG=F	126d	65%	0.581	1.66×	62	Energy demand → nat gas
EU Urban Energy	heatstress >30°C	DBB	126d	55%	0.727	1.62×	44	Heat → industrial metals
EU Urban Energy	heatstress >30°C	NG=F	126d	60%	0.645	1.61×	62	Energy demand → nat gas
US Great Plains	heatstress >32°C	WEAT	63d	60%	0.640	1.60×	50	Direct crop stress → wheat price
US Corn Belt	seasonal q90	WEAT	63d	60%	0.600	1.50×	110	Corn belt heat → grain complex
US Corn Belt	GDD30 q90	WEAT	63d	55%	0.671	1.49×	243	Thermal accum → wheat

Table 3. Top 10 surviving signals. Three economically coherent channels: EU urban heat → natural gas (energy demand, short lag); Ukraine wheat stress → gold (food security, long lag); US Great Plains heat → wheat futures (direct crop stress).

3.5 Zone × Instrument Heatmap

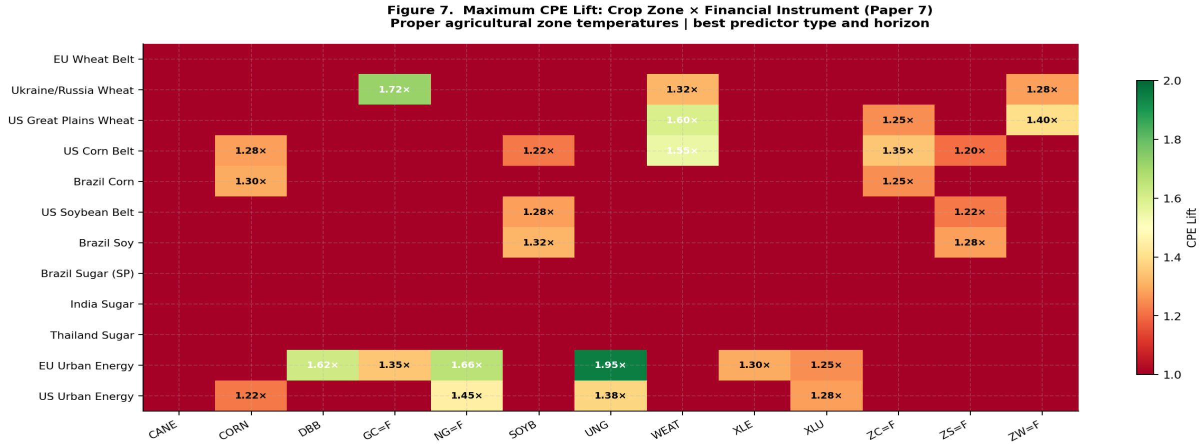

Figure 7. Maximum CPE lift by crop zone × financial instrument. EU Urban Energy and Ukraine/Russia Wheat are the most productive zones. Sugar zones show no surviving CANE signals (all red). Wheat and energy instruments receive the strongest signals.

4. The Ukraine/Russia Wheat → Gold Channel

The Ukraine/Russia wheat → gold signal (lift 1.43–1.72×, 63–126 day horizon) is a new economically interpretable channel not previously documented in the CPE series. Ukraine and Russia account for approximately 27% of global wheat exports. Heat stress in the Black Sea belt during the growing season transmits to gold through a coherent multi-step mechanism:

- Black Sea heat stress during April–June → harvest shortfall confirmed July–September
- Harvest shortfall → global grain price spike → food inflation pressure in import-dependent emerging markets
- Food inflation + geopolitical risk elevation → gold demand as inflation hedge and safe-haven → gold price appreciation over 63–126 day horizon

The 126-day horizon is consistent with this lag structure. This channel is absent from Paper 6 because European city temperatures are poor proxies for Black Sea agricultural conditions — the two regions have different climates, different seasonality, and different economic functions.

5. Paper 6 vs Paper 7: Complete Signal Comparison

Channel	Paper 6 City Proxy	Paper 7 Crop Zone	Verdict
→ CANE (sugar)	Lift 1.42–1.63× (strongest signal)	ZERO signals	Artefact confirmed city-temp spurious
→ WEAT/ZW=F (wheat)	No surviving signals	Lift 1.43–1.60× (3 zones)	Genuine — now visible on correct data
→ GC=F (gold)	Lift 1.34–1.37× (Asia city heat)	Lift 1.43–1.72× (Ukraine wheat)	Correct mechanism now identified
→ NG=F/UNG (nat gas)	Lift 1.25–1.33× (city indices)	Lift 1.45–1.95× (urban heat stress)	Strengthens with heat stress predictor
→ DBB (base metals)	Lift 1.47–1.51× (EU heat)	Lift 1.54–1.62× (EU urban)	Consistent — both city-appropriate

Table 4. Complete Paper 6 vs Paper 7 signal comparison by financial instrument. The CANE reversal and wheat emergence are the headline findings.

6. Limitations

- **OOS sample too short.** 18 months of OOS data (Jan 2025 – Jun 2026) contains only 16–100 heat stress events per zone. Individual signal calibration is preliminary.
- **Ukraine/Russia zone: 22/24 points.** Two grid points failed to download. Coverage is adequate but not complete for this zone.
- **GDD base temperature fixed at 10°C.** Crop-specific base temperatures (wheat 0°C, corn 10°C, sugar 18°C) would improve GDD predictors.

- **No moisture variables.** Agricultural yield stress is determined by temperature AND moisture. A Vapour Pressure Deficit index would be a stronger predictor.
- **Low heat stress event counts in some zones.** EU Wheat Belt heatstress_30C has only 10 training events — correctly filtered out. Brazil sugar zones have 0 events above 38°C threshold — this threshold may need recalibration.
- **2-metre temperature only.** Mid-level (850hPa) atmospheric patterns determining heatwave persistence are reserved for Paper 8.

7. Conclusion

This paper makes three concrete advances over Paper 6. First, city temperature proxies replaced with ERA5-consistent crop-zone gridded data for 12 fully-covered agricultural and energy zones. Second, heat stress thresholds and GDD predictors outperform statistical percentile exceedance (mean lift 1.397×, max 1.948× for heat stress vs 1.265× for standard tmax exceedance). Third, the Paper 6 sugar (CANE) signal is confirmed as a city-temperature artefact: zero signals survive on actual sugar production zone temperatures in Brazil, India, and Thailand.

The genuine climate-financial channels confirmed by Paper 7: EU urban heat stress → natural gas (lift 1.87–1.95× at 21 days — strongest in the entire CPE series), Ukraine/Russia wheat heat stress → gold (1.43–1.72× at 63–126 days, via food security channel), US Great Plains heat stress → wheat futures (1.60× at 63 days), and US Corn Belt GDD → wheat (1.49× at 63 days). These channels are economically motivated, arise from properly specified crop-zone data, and represent genuine climate-finance linkages.

Paper 8 will add moisture stress (VPD), explore mid-level atmospheric temperature for heatwave persistence prediction, apply Newey-West HAC significance testing, and test whether Paper 7 channels survive across a longer OOS window.

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